

THABISO KOJOANA

Projects

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Wits graduate with a BSc in Chemistry and BSc Honours focused on chemistry and computational chemistry. Currently awaiting final MSc Chemistry results. Independently developing programming skills since 2022, with a focus on Java, backend development and practical applications.

Technology stack

Languages & web: Java, JavaScript, SQL, HTML5, CSS3; responsive layouts, DOM events and Fetch API.

Backend & databases: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, REST APIs, Bean Validation, PostgreSQL and Flyway.

Android: Android SDK, Room, WorkManager, CameraX, ML Kit and Material Components.

Testing & development: JUnit, Mockito, MockMvc, Spring Boot Test, Testcontainers, Git, Maven, Gradle and AI-assisted development with code review and testing.

Cloud & delivery: Docker; AWS EC2, RDS, ECR, S3 and CloudFront; Caddy HTTPS proxy.

Honours: academic exposure: Computational chemistry, coding and data handling; supervised and unsupervised learning; neural-network concepts including convolutional (CNN) and recurrent (RNN) networks.

CAPSTONE PROJECT / INDEPENDENT BUILD

Kasi-Till

Android point-of-sale system and owner portal

Developed a mobile-first sales and stock-management system for South African spaza shops. An Android phone provides an affordable till during power interruptions, with local Room storage and queued synchronisation when connectivity returns.

Built Spring Boot REST APIs for products, staff access, sales and reporting; used PostgreSQL and Flyway for persistence and migrations. Implemented retry-safe uploads, ownership checks and licence handling, and deployed the Dockerised backend to AWS. Used JUnit, Mockito and controller tests to verify backend behaviour.

[Watch demonstration](#) | [Open owner portal](#)

Portal access requires an account. Private source walkthrough available on request.

Projects

Supporting projects | JavaScript, HTML and CSS

Five-Day Weather Studio

Built a responsive weather simulator with six South African city options, Celsius/Fahrenheit switching and refresh controls. Generated five-day cards show high and low temperatures, rain chances and wind speeds. Unit changes preserve the underlying scenario.

Demonstrates: arrays, state management, date handling, DOM rendering and responsive design. Values are simulated, not live weather data.

[Open demo](#) | [View source code](#)

Interactive CSS Robot

Created a robot from CSS shapes and JavaScript event handlers. The arm rotates 90 degrees at the shoulder and returns on a second click. Keyboard controls, reduced-motion support and responsive arm sizing make the interaction more accessible.

Demonstrates: event handling, CSS transforms, animation state and keyboard interaction.

[Open demo](#) | [View source code](#)

Dream Car Builder

Built a car customiser with editable make, model, year, body style, colour and price. Form inputs update a visual car preview and a readable JavaScript object. Validation rejects invalid entries, while a reset control restores the example.

Demonstrates: object modelling, form validation, safe text rendering and South African currency formatting.

[Open demo](#) | [View source code](#)

Name List Manager

Built an interactive list with add, remove, alphabetical sorting, reverse-order and reset controls. Blank or duplicate names are rejected, and names are displayed safely as text. The interface supports keyboard use and smaller screens.

Demonstrates: array operations, validation, DOM updates and accessible feedback.

[Open demo](#) | [View source code](#)

Train Animation Studio

Completed an interactive train demo with start, pause, adjustable speed, reverse, single-step and reset controls. The train turns around at track edges, stays within a responsive scene and pauses when the browser tab is hidden.

Demonstrates: frame-based animation, elapsed-time movement, boundary handling, state management and keyboard-accessible controls.

[Open demo](#) | [View source code](#)

These small web projects extend earlier learning exercises associated with JavaScript For Kids For Dummies. They complement Kasi-Till by showing focused programming and interface skills.